

# Graph Attention Networks for Multi-Isotope PET Random Events Correction

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## I. INTRODUCTION

Positron emission tomography (PET) is a powerful medical imaging technology that directly visualizes the body's metabolic processes, making it indispensable in the fields of Oncology, Neurology, and Cardiology. The clinical value of PET is entirely contingent on the quality of the image data it produces, making the correction of physical degradation effects a critical component of diagnostic reliability and patient outcome.

Random coincidences are a major source of image degradation in PET imaging, occurring when detectors falsely pair two unrelated photons as a single annihilation event. This problem is compounded in multi-isotope scans involving impure positron emitters like Mn-52, where an additional prompt gamma can be detected alongside the two annihilation photons: A so-called "triple coincidence" event. Current correction methods address this by rejecting such events outright.

This paper frames random event correction as a supervised binary classification task using a deep learning architecture composed of an encoder, Graph Attention (GAT) layers, and a multilayer perceptron (MLP) edge classifier. The model takes as input graphs encoding 15ns windows containing nodes of single detector hits simulated from GATE, a Monte Carlo simulation framework used to model particle transport and detector interactions in PET systems. Its output is a binary classification (True, Random) for every edge combination within the 15ns window.

## II. RELATED WORK

### A. Classical Analytical Methods

The two dominant approaches to random coincidence estimation are the delayed coincidence window (DCW) method and the singles-rate (SR) method. The DCW technique subtracts events captured in a time-shifted window from the prompt sinogram. This method is noisy, since the randoms estimate itself carries Poisson fluctuations. The SR method instead computes expected randoms per line-of-response analytically from individual detector singles rates. Oliver et al. [1] showed that SR systematically overestimates the true random rate, particularly at low energy thresholds, and proposed a corrected "Singles-Prompts" formulation that more accurately models pile-up.

### B. Deep Learning for PET Correction

A growing body of work applies neural networks to PET image quality problems. Reader et al. [2] organize deep learning methods for PET into three categories: post-processing denoising, direct sinogram-to-image reconstruction, and hybrid iterative methods. Sanaat et al. [3] showed that operating in sinogram space rather than image space yields superior PSNR and lower SUV bias for low-dose enhancement, which is a relevant finding given that our approach similarly acts close to the raw measurement data. An early contribution is Michaud et al. [4], who trained a feedforward neural network on Monte Carlo-simulated data to recover true lines-of-response from inter-crystal scatter triple coincidences in the LabPET scanner. By identifying the most probable LOR from each triplet rather than discarding it, their method achieved a 54% increase in scanner sensitivity. However, this paper focuses on prompt gamma emission in non-pure  $\beta^+$  emitters rather than inter-crystal scatter. No work has explicitly modeled the pairwise detector topology as a graph.

### C. Graph Attention Networks

This paper frames random coincidence correction as a problem on a detector graph, where nodes represent individual detector hits and edges represent lines-of-response. We employ GATv2 (Brody et al. [5]), which addresses the static attention limitation of the original GAT by introducing a dynamic attention mechanism. Critically, GATv2 is substantially more robust to edge noise. Unlike CNN-based methods, a graph network respects the topology of the coincidence system

## III. DATASET AND FEATURES

### A. Simulation Framework

All training and evaluation data were custom generated using GATE, a Monte Carlo simulation platform widely used for nuclear medicine imaging research due to its validated models of photon transport and detector response [7]. A cylindrical PET scanner geometry from PETcoil [9] was defined within the simulation environment, with the detector material and the timing resolution parameters matched to a representative system. Simulated acquisitions were configured to produce full list-mode output, retaining all single detector hits prior to coincidence sorting, thereby preserving the complete per-event information required for event-by-event ML processing. Within the detector ring, an acrylic cylinder filled with water

was placed to simulate the NEMA NU-2 count rate phantom [6].

### B. Dataset Composition

To ensure broad coverage of the operating conditions encountered in multi-isotope scans, a structured set of sub-datasets was generated spanning variations in isotope composition, relative activity, and spatial offset between isotope sources.

**Single-isotope acquisitions** were generated independently for  $^{18}\text{F}$  and  $^{52}\text{Mn}$ , providing clean single-species reference datasets for baseline model training and isotope-specific performance evaluation.  $^{18}\text{F}$  was selected as the primary clinical isotope of interest given its ubiquity in FDG-PET, while  $^{52}\text{Mn}$  was chosen as the secondary isotope due to its prompt-gamma emission profile. **Dual-isotope acquisitions** were generated across a factorial combination of simulation parameters.

### C. Feature Extraction

For each simulated acquisition, raw list-mode output was parsed to extract a per-event feature vector from every single detector hit, comprising the five-dimensional event  $(X, Y, Z, E, t)$ , where  $X, Y, Z$  denote the three-dimensional Cartesian interaction coordinates within the detector crystal,  $E$  denotes the deposited energy in keV, and  $t$  denotes the absolute detection timestamp in nanoseconds. No coincidence sorting was applied at the feature extraction stage.

To structure the list-mode single-hit data for processing by a graph neural network (GNN), a sliding temporal window of width 15 ns was applied to the chronologically ordered event stream. This window width was selected to span approximately twice the coincidence timing window of the simulated scanner, ensuring that all prompt coincidence pairs: true, scattered, and randoms are captured within a single window while limiting the number of uncorrelated background hits.

Each coincidence window is formalized as an undirected graph  $\mathcal{G} = (\mathcal{V}, \mathcal{E})$ , where the node set  $\mathcal{V}$  contains all photon interactions detected within the window and the edge set  $\mathcal{E}$  is fully connected. This construction allows the network to learn which pairs of detections constitute true coincidences. Each node  $v_i$  carries a feature vector

$$\mathbf{x}_i = [E_i, x_i, y_i, z_i, \hat{t}_i], \quad (1)$$

where  $E_i$  is the deposited energy,  $(x_i, y_i, z_i)$  is the interaction position, and  $\hat{t}_i \in [0, 1]$  is the arrival time normalized relative to the window width  $\Delta t$ ,

$$\hat{t}_i = \frac{t_i - t_{\min}}{\Delta t}. \quad (2)$$

Each edge  $(i, j)$  is assigned a binary label

$$y_{ij} = \mathbf{1}[\text{EventID}_i = \text{EventID}_j], \quad (3)$$

indicating whether the two interactions originate from the same annihilation event. The resulting graphs are stored using the PyTorch Geometric Data object, which encodes connectivity in coordinate (COO) sparse format as an edge index tensor  $\mathbf{E} \in \mathbb{Z}^{2 \times |\mathcal{E}|}$ , where  $|\mathcal{E}| = \binom{|\mathcal{V}|}{2}$  for a fully connected graph.

### D. Simulation verification

To validate the simulation environment and dataset composition, the energy distribution of singles was examined for both F-18 and Mn-52 sources. For both isotopes, a photopeak at 511 keV is expected, confirming the presence of proper positron annihilation events. The water-filled cylindrical phantom introduces a substantial population of low-energy singles attributable to Compton scattering interactions within the medium. In addition to the 511 keV annihilation peak, Mn-52 is expected to exhibit prominent photopeaks at 1434 keV, 935 keV, and 744 keV, corresponding to its characteristic prompt gamma emissions.

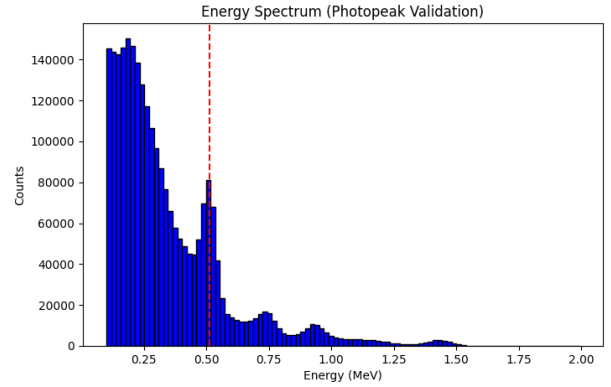


Fig. 1. Examination of a sub-dataset containing both  $^{18}\text{F}$  and  $^{52}\text{Mn}$  line sources. The energy distribution shows a clear peak at 511 keV, followed by subpeaks at approximately 744, 935, and 1434 keV. A signification portion of signals are also low-energy compton scatters. This energy distribution verifies the data acquisition framework

### E. Final Dataset Metrics

The final dataset comprised 250,000 graphs drawn from each subdataset, shuffled to yield a total of 3,250,000 graphs. These were partitioned into training, validation, and testing splits of 70%, 15%, and 15%, respectively. To mitigate class imbalance, positive examples were upweighted such that the number of edges classified as random was equal to the number of edges classified as true.

## IV. METHODS

### A. Model Architecture

The proposed model, PET\_Randoms\_GNN, is a graph neural network consisting of three stages: a node feature encoder, a stack of graph attention layers for message passing, and an MLP-based edge classifier.

1) *Node Encoder*: Raw node features are first projected into a  $d$ -dimensional latent space via a learned linear transformation,

$$\mathbf{h}_i^{(0)} = \text{ReLU}(\mathbf{W}_{\text{enc}} \mathbf{x}_i + \mathbf{b}_{\text{enc}}), \quad (4)$$

where  $\mathbf{W}_{\text{enc}} \in \mathbb{R}^{d \times 5}$  and  $\mathbf{b}_{\text{enc}} \in \mathbb{R}^d$ , with hidden dimension  $d = 64$ .

2) *Graph Attention Layers (GATv2)*: Node representations are refined through  $L$  successive GATv2 message-passing layers [5]. Unlike the original GAT, GATv2 computes a *dynamic* attention coefficient that is a function of both the source and target node representations, thereby resolving the static attention problem. For a node  $i$  and its neighbour  $j$ , the unnormalised attention score at layer  $\ell$  is

$$e_{ij}^{(\ell)} = \mathbf{a}^\top \text{LeakyReLU}\left(\mathbf{W} \left[ \mathbf{h}_i^{(\ell)} \parallel \mathbf{h}_j^{(\ell)} \right]\right), \quad (5)$$

where  $\mathbf{W} \in \mathbb{R}^{d' \times 2d}$  is a learnable weight matrix,  $\mathbf{a} \in \mathbb{R}^{d'}$  is a learnable attention vector, and  $\parallel$  denotes concatenation. The scores are normalised over all neighbours  $\mathcal{N}(i)$  via softmax,

$$\alpha_{ij}^{(\ell)} = \frac{\exp(e_{ij}^{(\ell)})}{\sum_{k \in \mathcal{N}(i)} \exp(e_{ik}^{(\ell)})}. \quad (6)$$

Multi-head attention with  $K$  heads is employed to stabilise learning. Each head  $k$  computes an independent set of coefficients  $\alpha_{ij}^{(\ell,k)}$ , and the resulting representations are concatenated to form the updated node embedding,

$$\mathbf{h}_i^{(\ell+1)} = \text{ELU}\left(\left\|_{k=1}^K \sum_{j \in \mathcal{N}(i)} \alpha_{ij}^{(\ell,k)} \mathbf{W}^{(k)} \mathbf{h}_j^{(\ell)}\right.\right), \quad (7)$$

where  $\left\|_{k=1}^K$  denotes concatenation across all  $K$  heads. With  $K = 4$  heads and a per-head output dimension of  $d/K = 16$ , the full concatenated representation has dimension  $d = 64$  throughout all  $L = 2$  layers.

3) *Edge Classifier*: Following message passing, physics-informed edge features are constructed by combining the context-enriched node representations with hand-crafted interaction terms. For a candidate pair  $(i, j)$ , the edge feature vector is

$$\mathbf{e}_{ij} = [\mathbf{h}_i \parallel \mathbf{h}_j \parallel |\Delta t_{ij}| \parallel \|\Delta \mathbf{r}_{ij}\|_2] \in \mathbb{R}^{130}, \quad (8)$$

where

$$|\Delta t_{ij}| = |t_i - t_j|, \quad (9)$$

$$\|\Delta \mathbf{r}_{ij}\|_2 = \sqrt{(X_i - X_j)^2 + (Y_i - Y_j)^2 + (Z_i - Z_j)^2} \quad (10)$$

are the absolute time difference and Euclidean crystal separation, respectively. These physics-motivated quantities provide the classifier with direct access to the time-of-flight and spatial information most discriminating for random rejection.

The edge feature vector is passed through a three-layer MLP,

$$\hat{y}_{ij} = \mathbf{W}_3 \text{ReLU}(\mathbf{W}_2 \text{ReLU}(\mathbf{W}_1 \mathbf{e}_{ij} + \mathbf{b}_1) + \mathbf{b}_2) + b_3, \quad (11)$$

with layer widths  $130 \rightarrow 128 \rightarrow 64 \rightarrow 1$  and dropout with probability  $p = 0.2$  applied after each hidden activation. The scalar output  $\hat{y}_{ij}$  is interpreted as the logit for the true-coincidence class.

## B. Advantages of the Proposed Approach

1) *Per-Coincidence Classification*: The proposed model instead performs a direct, per-coincidence binary classification. For each candidate pair  $(i, j)$ , the network produces a logit  $\hat{y}_{ij}$  that reflects the probability of that specific coincidence being true, allowing individual events to be accepted or rejected prior to reconstruction. This avoids the noise amplification inherent to statistical subtraction and enables a cleaner listmode dataset.

2) *Multi-Hit Contextual Reasoning*: A fundamental limitation of methods that operate on coincidence pairs in isolation, including both conventional techniques and naive MLP-based classifiers, is that each decision is made without any knowledge of the broader acquisition context. In a realistic listmode acquisition window, a single detector hit may participate in several candidate coincidence pairs simultaneously. The validity of any one pairing is therefore not independent of the others.

## C. Limitations of the Proposed Approach

1) *Computational Cost*: GNNs are more computationally demanding than conventional randoms correction methods. The DCW technique, for instance, requires only a simple sinogram subtraction and adds negligible overhead to the reconstruction pipeline. In contrast, the proposed model must construct a graph for each acquisition window, perform  $L$  rounds of attention-based message passing, and evaluate the edge classifier for every candidate pair. At high count rates, this may present a bottleneck for clinical deployment where near-real-time processing is expected.

2) *Training Data Requirements*: Supervised training requires a large corpus of labelled listmode data in which individual coincidences are known to be true or random. In simulation this is straightforward, but obtaining such ground-truth labels from real scanner acquisitions is not possible without auxiliary information. The model's performance is ultimately bounded by the fidelity of the simulation used to generate training data. float stfloats

## V. EXPERIMENTS

In PET acquisition, the random coincidence rate scales approximately with the square of the activity concentration, such that at low activity levels true coincidences substantially outnumber randoms, while at high activity levels the two classes approach balance. This inherent variability in class balance across acquisition conditions requires a metric beyond accuracy as a reliable performance metric. Consequently, this work adopts the F1 score and the  $F_{0.5}$  score as the primary evaluation metrics.

The  $F_{0.5}$  score is of particular interest, as it weights precision twice as heavily as recall:

$$F_{0.5} = \frac{(1 + 0.5^2) \cdot \text{Precision} \cdot \text{Recall}}{0.5^2 \cdot \text{Precision} + \text{Recall}} \quad (12)$$

This weighting reflects the operational objective of the model: to pass the cleanest possible set of true coincidences to the downstream reconstruction pipeline. In this context, falsely

retaining a random coincidence (a false positive) is considered more detrimental than discarding a true coincidence (a false negative). Precision is therefore the quantity to be maximised, and  $F_{0.5}$  provides a principled metric that encodes this priority.

#### A. Hyperparameter Selection

The model was trained using stochastic gradient descent with a learning rate of  $\eta = 2.5 \times 10^{-3}$  and a mini-batch size of 256 graphs. These values were selected empirically through preliminary training runs, in which the evolution of the training loss was monitored to identify a configuration that achieved stable convergence without divergence. Hardware memory constraints imposed by the available GPU were also taken into account when determining the batch size.

Architectural hyperparameters were selected via grid search, where each candidate configuration was trained for 30 epochs and evaluated on the validation set using the  $F_{0.5}$  score as the selection criterion. The hyperparameters tuned and the values searched are summarized in Table I, with the selected configuration yielding

TABLE I

HYPERPARAMETER CANDIDATES EVALUATED DURING GRID SEARCH AND THE SELECTED VALUES. SELECTION WAS BASED ON VALIDATION  $F_{0.5}$  SCORE AFTER 30 TRAINING EPOCHS.

| Hyperparameter       | Candidates   | Selected |
|----------------------|--------------|----------|
| Hidden dimension $d$ | 64, 128, 256 | 64       |
| GATv2 layers $L$     | 1, 2, 3      | 2        |
| Attention heads $K$  | 4, 8         | 4        |

#### B. Performance

The model achieving the highest  $F_{0.5}$  score after an initial 30-epoch search was selected for full training over 150 epochs. Training and validation loss,  $F_1$ , and  $F_{0.5}$  converged at approximately 100 epochs (Fig. 2). Final performance metrics on the test dataset at a classification threshold of 0.5 are reported in Table II.

TABLE II

PERFORMANCE METRICS OF GNN MODEL ON TEST DATASET (THRESHOLD = 0.5)

| Metric    | Value  |
|-----------|--------|
| Accuracy  | 0.9479 |
| Precision | 0.8575 |
| Recall    | 0.9887 |
| $F_1$     | 0.9184 |
| $F_{0.5}$ | 0.8809 |

The persistently higher  $F_1$  relative to  $F_{0.5}$  indicates a recall-biased model. With recall approaching 0.989 at the default threshold, a higher threshold is warranted to improve precision.

#### C. Error Analysis

A rudimentary scatter correction is applied by imposing an energy floor of 350 keV; coincidence pairs in which either photon falls below this threshold are excluded from further analysis. Inspection of the false positive energy distribution

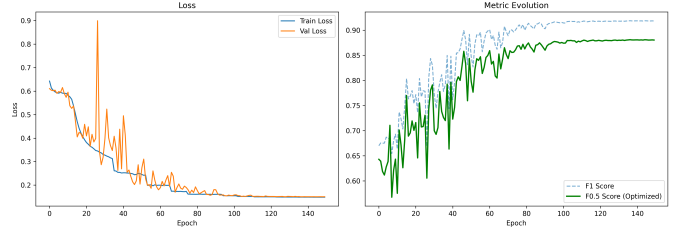


Fig. 2. Training and validation loss (left) and  $F_1/F_{0.5}$  scores (right) over 150 epochs. Convergence occurs at approximately 100 epochs.

(Fig. 3) reveals that a substantial fraction of false positives cluster at low energies, confirming that this threshold effectively removes a large class of misclassified scatter events.

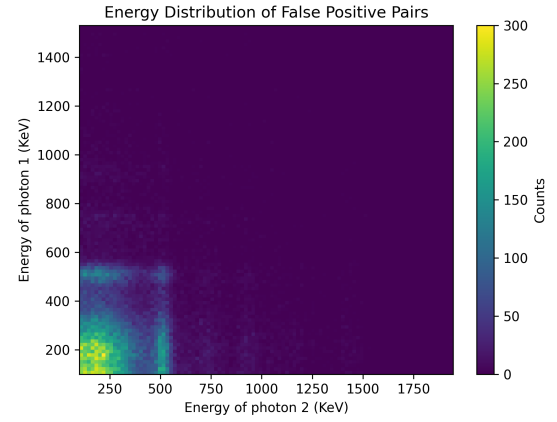


Fig. 3. 2D histogram of false positive event energies. The concentration of errors at low energies corresponds to scatter events removed by the 350 keV energy threshold.

Since false positives are more detrimental to PET image quality than false negatives, the output threshold was tuned to 0.858, the value yielding maximum precision at  $\geq 90\%$  recall (Fig. 4). Applying both the energy threshold and the tuned classification threshold, the resulting confusion matrix is shown in Fig. 5.

TABLE III

PERFORMANCE METRICS OF GNN MODEL ON TEST DATASET (THRESHOLD = 0.858)

| Metric    | Value    |
|-----------|----------|
| Accuracy  | 0.961184 |
| Precision | 0.925885 |
| Recall    | 0.929038 |
| $F_1$     | 0.927459 |
| $F_{0.5}$ | 0.926514 |

#### D. Threshold Tuning

A practical advantage of the proposed model is the ability to adjust the output threshold to suit different imaging regimes. Low-dose acquisitions benefit from maximizing recall to retain all true events, while high-activity scans demand higher

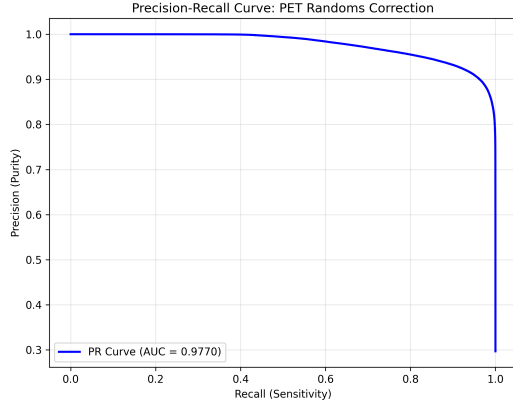


Fig. 4. Precision-recall curve. A threshold of 0.858 achieves  $F_{0.5}$  at 90% recall.

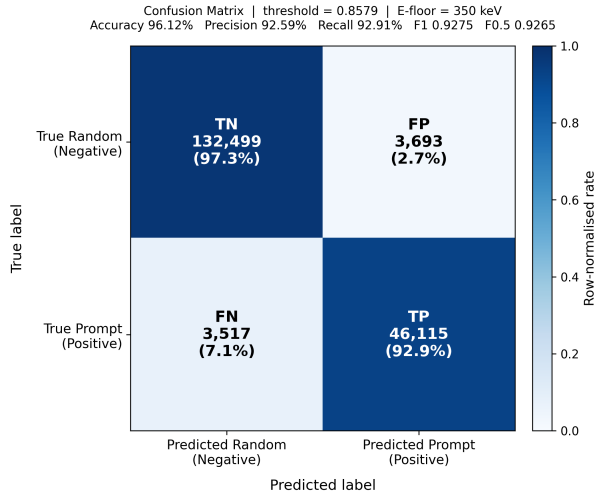


Fig. 5. Confusion matrix after applying the 350 keV energy floor and classification threshold of 0.858.

precision to suppress randoms. Table IV summarizes model performance across a range of thresholds, demonstrating a smooth and predictable precision-recall trade-off.

TABLE IV  
GNN PERFORMANCE ACROSS OUTPUT THRESHOLDS

| Threshold | $F_{0.5}$ | Precision | Recall |
|-----------|-----------|-----------|--------|
| 0.50      | 0.8688    | 0.8413    | 0.9993 |
| 0.60      | 0.8843    | 0.8598    | 0.9982 |
| 0.70      | 0.9029    | 0.8827    | 0.9937 |
| 0.80      | 0.9203    | 0.9083    | 0.9718 |
| 0.90      | 0.9224    | 0.9435    | 0.8465 |
| 0.95      | 0.8656    | 0.9688    | 0.6070 |

## VI. CONCLUSION

The proposed GATv2-based model demonstrates strong performance on the random coincidence classification task, achieving an  $F_{0.5}$  score of 0.9265 - optimal threshold of  $>0.90$

recall. The confusion matrix confirms that the vast majority of events are correctly classified, with errors representing a small fraction of total coincidences. The tunable output threshold is a meaningful clinical feature: it allows the same trained model to be adapted to low-dose or high-activity acquisition protocols without retraining, offering flexibility unavailable in fixed analytical corrections such as the delayed coincidence window or singles-rate methods.

Several limitations merit acknowledgment. The model is trained and evaluated entirely on simulated data, and generalization to real scanner geometries, detector non-uniformity, and electronic noise remains to be validated. The scatter correction applied here is rudimentary, and a more principled treatment may further reduce false positives independent of the classifier. Future work should evaluate performance on experimental phantom and clinical data and test the model's generalization to different PET machines.

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## REFERENCES

- [1] J. F. Oliver and M. Rafecas, "Singles-prompts-randoms: Estimation of spurious data rates in PET," in Proc. IEEE Nucl. Sci. Symp. Med. Imag. Conf., Anaheim, CA, USA, Oct. 2012, pp. 3667–3670.
- [2] A. J. Reader, G. Corda, A. Mehranian, C. da Costa-Luis, S. Ellis, and J. A. Schnabel, "Deep learning for PET image reconstruction," IEEE Trans. Radiat. Plasma Med. Sci., vol. 5, no. 1, pp. 1–25, Jan. 2021.
- [3] A. Sanaat, H. Arabi, I. Mainta, V. Garibotto, and H. Zaidi, "Projection space implementation of deep learning-guided low-dose brain PET imaging improves performance over implementation in image space," J. Nucl. Med., vol. 61, no. 9, pp. 1386–1396, Sep. 2020.
- [4] J.-B. Michaud, C.-A. Brunet, M. Rafecas, R. Lecomte, and R. Fontaine, "Sensitivity increase through a neural network method for LOR recovery of ICS triple coincidences in high-resolution pixelated-detector PET scanners," IEEE Trans. Nucl. Sci., vol. 62, no. 1, pp. 52–62, Feb. 2015.
- [5] S. Brody, U. Alon, and E. Yahav, "How attentive are graph attention networks?" in Proc. Int. Conf. Learn. Represent. (ICLR), Apr. 2022. [Online]. Available: <https://openreview.net/forum?id=F72xmsx7C1>
- [6] R. E. Schmitz, R. L. Harrison, C. W. Stearns, T. K. Lewellen, and P. E. Kinahan, "Optimization of noise equivalent count rate performance for a partially collimated PET scanner by varying the number of septa," IEEE Trans. Med. Imag., vol. 26, no. 7, pp. 935–944, Jul. 2007.
- [7] GATE (the canonical reference is the 2004 OpenGATE collaboration paper in Physics in Medicine and Biology): S. Jan, G. Santin, D. Strul, S. Staelens, K. Assié, D. Autret, S. Avner, R. Barbier, M. Bardiès et al., "GATE: A simulation toolkit for PET and SPECT," Phys. Med. Biol., vol. 49, no. 19, pp. 4543–4561, 2004.
- [8] Matplotlib: J. D. Hunter, "Matplotlib: A 2D graphics environment," Comput. Sci. Eng., vol. 9, no. 3, pp. 90–95, May 2007, doi: 10.1109/MCSE.2007.55.
- [9] Q. Dong, M. N. Ullah, D. Innes, R. D. Watkins, C.-M. Chang, S. J. Zou, A. Groll, I. Sacco, G. Chinn, and C. S. Levin, "PETcoil: First results from a second-generation RF-penetrable TOF-PET brain insert for simultaneous PET/MRI," Phys. Med. Biol., vol. 69, no. 18, Sep. 2024, doi: 10.1088/1361-6560/ad7221.